

MT_PBR STUDIO PRO

1-Click Architectural PBR Material & Lighting Solution for Blender

1-Click PBR Maps & Fabric Micro-Sheen

Converts any single diffuse texture into a complete 3D relief material; Micro-Sheen technology completely eliminates synthetic plastic sheen on sofa fabrics and leather cushions.

Visual Surface Imperfections & Scratches

Translate and rotate surface dirt/scratches directly on 3D Viewport using intuitive mouse gizmos (G/R); smart slope mask keeps cabinet tops clean.

Backlit Translucent Stone (Onyx)

Creates glowing TV feature walls and bar counters; Invert mode switches seamlessly between illuminated body or glowing stone veins.

Wood Grain Align & Quick Box UVW

1-Click wood grain vertical/horizontal alignment; 6-direction Triplanar Box mapping without UV unwrapping; Ultra-lightweight anti-freeze viewport HDRI.

DEVELOPED BY

MT Studio • MT_Hub Ecosystem

Professional Architectural 3D & Graphics Add-on Suite

Version: v3.12.0 Pro • Compatibility: Blender 4.0 - 5.3+

Standard English Edition

Chapter 1: Quick Start & MT_Hub Master Console

1.1. 1-Click Installation into Blender

- 1 Open Blender, navigate to **Edit** > **Preferences...** > **Add-ons**.
- 2 Click **Install...**, select the **MT_PBR_Studio.zip** file, and check the checkbox to activate.
- 3 In the 3D Viewport, press key **N** to open the Sidebar panel and switch to tab **MT_PBR Studio Pro**.

1.2. Two-Way MT_Hub Master Ecosystem Link

MT_PBR Studio Pro links directly to the **MT_Hub Master** console:

- **Automatic License Verification:** Recognizes lifetime activation or 5-day trial instantly without re-entering keys.
- **1-Click Online Updates:** Cloud update notifications appear automatically when new versions are released.
- **Real-Time Bilingual Sync:** When you toggle between English and Vietnamese on either tab, both add-ons synchronize immediately.

1.3. 8 Specialized Filter Tabs

Filter your workspace efficiently using the 8 quick tabs at the top of the panel:

-  **ALL:** Show entire panel
-  **PBR:** 50 Materials & Eyedropper Cloner
-  **MAPS:** 1-Click PBR Map Generator
-  **DIRT:** Scratches, Dust & Geometric Dirt/AO
-  **UVW:** Wood Grain & Box Mapping
-  **HDRI:** 360° Lighting & Anti-Freeze Viewport
-  **DISP:** Hybrid 3D Displacement
-  **EXPORT:** Texture Packing & Game Export



Chapter 2: 50 Architectural PBR Presets & Eyedropper Cloner

MT_PBR Studio Pro provides an expansive built-in library of **50 standard architectural PBR materials** pre-calibrated with physically accurate optical parameters (IOR, roughness, metallic, sheen, subsurface):

Material Family	Key Presets in Library	Architectural Applications
 1. Wood (WOOD)	High-gloss Acrylic, Melamine, Matte PU, Natural Oak, Rough Sawn...	Acrylic kitchen cabinets, parquet flooring, dining tables, wall slats.
 2. Metal (METAL)	Perfect Mirror, Chrome, Hairline Brushed Inox, 24K Gold, Copper...	Bathroom mirrors, PVD gold trims, door handles, shower fittings.
 3. Stone & Tile (STONE)	Glossy Marble, Cloud Granite, Matte Ceramic, Flamed Pavers...	White porcelain countertops, pavement tiles, feature wall claddings.
 4. Paint (PAINT)	Matte Wall Paint, Satin Finish, Metallic Car Paint, Solid Pastel...	Interior drywall paint, MDF spray-painted cabinet doors.
 5. Glass (GLASS)	Clear Glass, Frosted Sandblasted, Smoked Black, Fluted Ribbed...	Shower enclosures, aluminum windows, luxury ribbed wardrobe doors.
 6. Procedural Water (WATER)	Sunny Pool Ripples, Ocean Waves, Stream, Calm Lake Surface...	Villa swimming pools, landscaping fountains, aquariums.
 7. Plastic & Ceramic (PLASTIC)	Sanitary Ceramic, ABS Plastic, Damping Rubber...	Porcelain toilets, electronic appliances, interior plastic accents.
 8. Fabric & Leather (FABRIC)	Glossy Leather, Suede, Rough Canvas, Velvet, Silk Drapes...	Luxury sofas, leather headboards, double-layer blackout curtains.
 9. Concrete (CONCRETE)	Polished Concrete, Exposed Formwork, Drywall Plaster...	Industrial floors, exposed concrete accent walls, ceiling gypsum.
 10. Emissive & Backlit Stone	Warm LED 3000K, Neutral 4000K, Cool 6500K, Backlit Onyx Stone ...	Cove ceiling lights, signage, luxury illuminated bar counters.

2.1. Rapid Eyedropper Material Cloner

Instead of manually setting parameters across multiple objects, use the **Eyedropper Cloner**:

 **Usage:** Select the target object -> In the  **PBR LIBRARY** section, click the eyedropper icon  **Eyedropper** -> Click on any reference object in the 3D scene. The entire material shader tree, textures, displacement modifiers, and effects are copied instantly!

2.2. "Make Single-User Copy" Option (Prevent Accidental Overwrites)

When cloning materials onto multiple objects, checking **Independent Material Copy** guarantees that editing parameters on one object will not inadvertently alter the others.

⚡ Chapter 3: 1-Click PBR Map Generator & Fabric Micro-Sheen

3.1. Common Issues with Web Textures

Most downloaded textures online are flat 8-bit RGB color images lacking Normal, Roughness, or AO maps. Rendering them results in artificial flatness and fake reflections.

3.2. 1-Click Complete PBR Map Generator

MT_PBR Studio Pro analyzes your single Base Color image algorithmically and automatically generates all 4 crucial architectural maps:

- **Normal Map (Sobel Algorithm):** Extracts directional micro-surface angles for crisp 3D lighting.
- **Micro-Relief Bump:** Adds fine tactile depth without geometry subdivision.
- **Calibrated Roughness Map:** Distributes gloss and matte zones naturally across texture crevices.
- **Cavity Ambient Occlusion (AO):** Bakes deep shadow crevices in grooved and textured surfaces.

3.3. Fabric & Leather Micro-Sheen (Zero Plastic Sheen)

Synthetic plastic reflections on sofas and cushions are completely eliminated through physical **Micro-Sheen** optical grazing calculation:

💡 What is Micro-Sheen?

Micro-Sheen simulates millions of microscopic fuzz fibers catching grazing light on fabric edges, providing soft velvety sheen at glancing angles while keeping facing surfaces natural and matte.



Chapter 4: Surface Imperfections, Scratches & Alpha Controls

4.1. 3 Visual Alpha Calibration Sliders

Fine-tune surface dirt, water drip marks, and finger scratches with absolute precision:

Slider Property	Function Description	Recommended Practical Values
Coverage Density (Density)	Filters faint dust haze; isolates distinct scratch marks.	0.00 (rich haze) to 0.35 (clean surfaces).
B/W Contrast (Contrast)	Sharpens edge separation between dirt and clean areas.	0.00 (smooth blend) to 0.50 (crisp edge grime).
Alpha Brightness (Brightness)	Scales the overall intensity and visibility of the imperfection layer.	-0.20 (subtle gloss) to +0.30 (heavy weathering).

4.2. Smart Slope Filter: Keeping Cabinet Tops Clean

Toggle **Clean Upward Faces** to automatically exclude vertical dust from horizontal countertops and table tops, mimicking real-world gravity-driven dirt accumulation.

4.3. Interactive Mouse Transform Gizmo (Modal G / R)

Click **Transform Gizmo** in the 3D Viewport:

- Press **G**: Move the scratch/dirt map across the surface with mouse.
- Press **R**: Rotate the scratch orientation seamlessly.
- Press **S**: Scale the imperfection frequency up or down.
- Click **Left Mouse**: Confirm new position • **Esc**: Cancel.

Chapter 5: Backlit Translucent Stone & Emissive Lighting

5.1. Luxury Interior Applications

Backlit Onyx and Translucent Stone are staples of ultra-luxury interior architecture, widely specified for illuminated bar counters, hotel reception desks, and glowing TV feature walls.

5.2. 1-Click Backlit Stone Generation from Any Image

Load any high-resolution marble or onyx texture: MT_PBR Studio Pro automatically splits luminance channels, routes Base Color to Diffuse, and feeds the extracted optical transparency mask to the Principled BSDF Emission shader.

5.3. Flexible Invert Mode (Glowing Veins vs. Glowing Body)

Invert Mode Options

- **Veins Glow:** Dense stone veins radiate amber/golden light while the main stone slab remains darker.
- **Body Glow:** The main crystal matrix glows brightly while dark mineral veins cast realistic deep silhouettes.

5.4. LED Glow Color Tinting & Alpha Preview Mode

Adjust the backlighting color (Warm 3000K, Neutral 4000K, Amber Gold) and toggle  **Preview Alpha** to inspect the black-and-white luminosity mask in real time.



Chapter 6: Wood Grain Align, Box UVW & Anti-Freeze HDRI

6.1. Smart 1-Click Wood Grain Alignment

Click  **Align Grain** to rotate texture orientation by 90° instantly, aligning grain along long boards, cabinet columns, or table edges without manual UV editing.

6.2. Triplanar Box UVW Mapping (No UV Unwrap Required)

Project seamless textures onto complex imported CAD/BIM geometries from 6 spatial directions simultaneously without seam distortion.

6.3. 360° HDRI Studio & Anti-Freeze Lightweight Viewport Mode

Rotate sunlight angle freely with the HDRI rotation dial. When working with massive architectural scenes, activate  **Lightweight Viewport Mode** to save **99% VRAM** and eliminate UI lag completely.



Chapter 7: Texture Packaging, Game Export & Shortcuts

7.1. 1-Click Texture Packaging (Render Farm Ready)

Click **Collect & Bundle Textures**: All image maps, HDRIs, and alpha masks linked across the blend file are automatically copied into a local `//textures/` folder with relative paths, preventing missing files when transferring to render farms.

7.2. PBR Map Export for Unreal Engine, Unity & 3ds Max

Bake and export standardized 16-bit PBR image sets (``_BaseColor``, ``_Normal``, ``_Roughness``, ``_Displacement``, ``_AO``) formatted for external DCC and game engine pipelines.

7.3. Handy Keyboard Shortcut Reference

Shortcut	Action Context	Description
N	3D Viewport	Open Sidebar Panel > Switch to Tab MT_PBR Studio Pro / MT Hub.
G / R / S	Modal Transform Gizmo	Translate, Rotate, or Scale dirt/scratches on viewport surfaces.
Ctrl + Z	Every Operator	Full undo support across all MT_PBR material actions.
Z > Rendered	3D Viewport Shading	Toggle interactive Cycles / EEVEE Next real-time preview.

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